

Tri-plane diagrams and surface knot invariants

Isaiah Martinez, Wolfgang Allred, Manuel Aragon, Zack Dooley, Alec Goldman, Yucong Lei, Nicholas Meyer, Devon Peters, Scott Warrander, Ana Wright, and Alexander Zupan

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Introduction

What is a knot?

Knots are mathematical objects depicted as one dimensional (closed) strands that live in our three-dimensional space. We like to imagine these knots are made up of *very* elastic rubber and that they have no width. Below is a diagram of the simplest knot, known as the unknot.

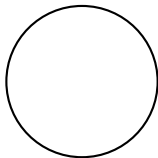


Figure: An unknot

Links

At times, it may be convenient to consider multiple knots “linked” together. Such a construction is so suitably named, a **link**. Here are a couple of the simplest known links.

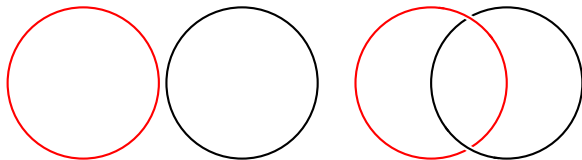


Figure: A two component unlink and a Hopf link (left to right)

How can we tell if two knots are the same?

Believe it or not, the picture below is also an unknot.

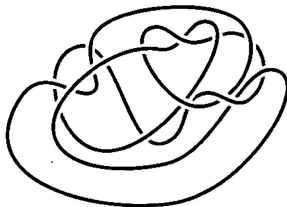


Figure: A tangled up unknot (Adams [1], Exercise 1.3)

How can we tell if two knots are the same?

Can you tell if these two knots are distinct?

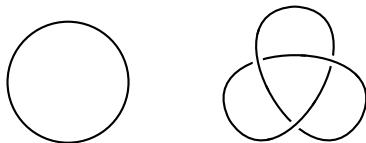


Figure: The unknot and the trefoil knot (left to right)

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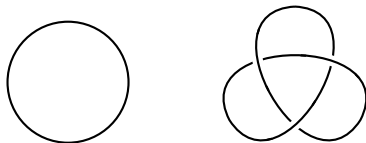


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It turns out that to prove these two knots are distinct, we need some more sophisticated machinery. Before we introduce such concepts, we are going to go up in dimension.

Surface Knots

Knots in Four-Space

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What we are interested in studying though are two-dimensional knots, which we will call **surface knots** (or **knotted surfaces**), in four-space. We can depict these surface knots with three planar diagrams, which we call **tri-plane diagrams**.

Tri-plane Diagrams

Definition

A **tri-plane diagram** $K = (K_1, K_2, K_3)$ must satisfy the following.

- Each diagram should have the same number of strands (and hence same number of endpoints on the horizontal axis).
- Each strand should be a trivial tangle, that is, each strand has one relative maximum with respect to the horizontal axis.
- Each of $K_1 \cup \overline{K_2}$, $K_2 \cup \overline{K_3}$, and $K_3 \cup \overline{K_1}$ should be diagrams for an unlink (where $\overline{K_i}$ is the mirror over the horizontal axis).



Figure: An example of a tri-plane diagram $K = (K_1, K_2, K_3)$.

Tri-plane Diagrams

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Clearly, each tangle is trivial and each diagram has four strands. Inspecting the below figure allows us to verify that each mirrored union is an unlink.

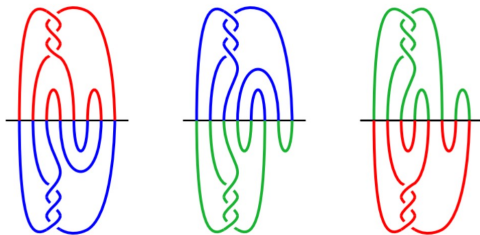


Figure: A verification that the mirrored unions are unlinks.

Tri-plane Moves

Just as in 3-dimensional knot theory, tri-plane diagrams can undergo local changes via the three Reidemeister moves.

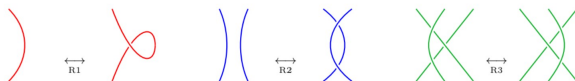


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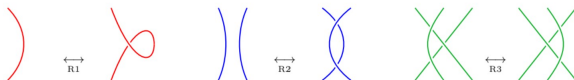


Figure: The three Reidemeister moves

Additionally, we can perform mutual braid moves, which are shown below

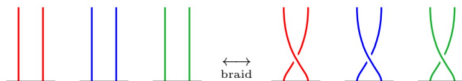


Figure: A mutual Braid Move

Tri-plane Moves

We can also perform stabilization and destabilization moves as pictured below.

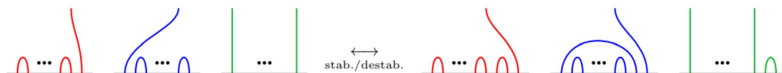


Figure: Stabilization and Destabilization

Tri-plane Moves

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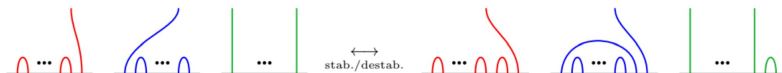


Figure: Stabilization and Destabilization

Definition

A **tri-plane move** is a manipulation of a tri-plane diagram via a Reidemeister move, a mutual braid move, or a stabilization/destabilization.

Knotted Surfaces

Definition

We write $K \sim K'$ if K' can be obtained from K via a finite series of tri-plane moves and planar isotopies. One can check that $\sim$ as defined above is an equivalence relation with equivalence class $[K]$. We call the equivalence class $[K]$ a **knotted surface**.

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Definition

We say that a surface **unknotted** if it is either the unknotted two-sphere U , or a finite connected sum of copies of the positive unknotted projective plane P^+ , the negative unknotted projective plane P^- , or the unknotted torus, T . We denote n connected sums of P^+ and m copies of P^- by $P^{n,m}$.

Connected Sums

But what is a connected sum?

Definition

Consider two tangle diagrams K_1 and K_2 . We can take a connected sum $K_1 \# K_2$ of the diagrams by detaching an endpoint on each diagram from the y -axis and connecting them. See the below figure.

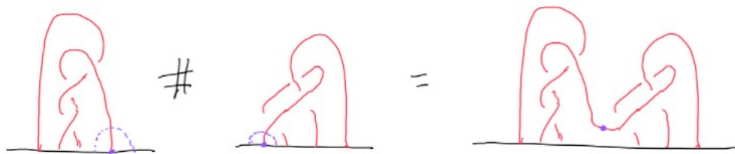


Figure: Connected sum of two tangle diagrams

Tri-plane Diagrams for the Simplest Unknotted Surfaces

Here are tri-plane diagrams for the simplest knotted surfaces.



(A) U



(B) P^+



(C) P^-



(D) T

Figure: (A) The unknotted two-sphere, (B) The positive unknotted projective plane, (C) The negative unknotted projective plane, (D) The unknotted torus

Orientability

Definition

Given a tri-plane diagram K with b strands, to each endpoint we will associate either a $+$ or a $-$. We do so in such a way so that each strand has a $+$ at one endpoint and a $-$ at the other. We also ensure that the signs are consistent between all three diagrams. If all $2b$ signs in each diagram match and each strand receives one sign each, then we say that K is **orientable**, and we write $o(K) = 1$. Otherwise, $o(K) = 0$.

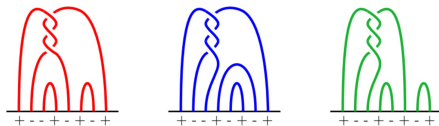


Figure: 7. The spun trefoil is orientable

Other Known Invariants

Definition

We define the **Euler characteristic** of a surface K as $\chi(K) = p_{12} + p_{23} + p_{31} - b$, where p_{ij} are the patch numbers and b is the bridge number.

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We define the **normal Euler number** of a surface K as $e(K) = w(K_1 \cup \overline{K_2}) + w(K_2 \cup \overline{K_3}) + w(K_3 \cup \overline{K_1})$.

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Orientability, Euler characteristic, and normal Euler number are all *surface invariants*. That is, if $K \sim K'$ then $o(K) = o(K')$, $\chi(K) = \chi(K')$, and $e(K) = e(K')$.

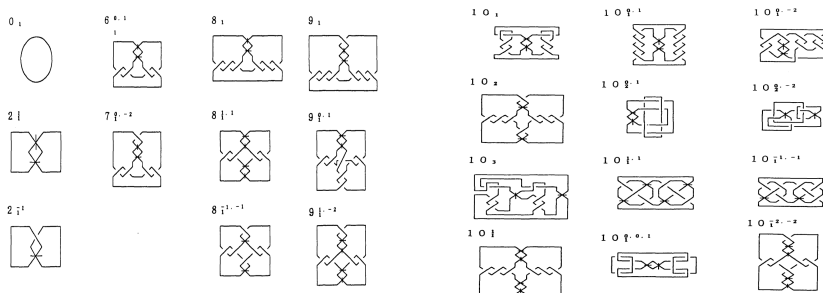
Compiling a List of Known knotted surfaces

CH-Diagrams

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CH-diagrams for surface knots (Yoshikawa [3])

Resolving Crossings

In order more easily compute invariants, we turn the CH-diagrams into tri-plane diagrams. In order to do so, we must resolve these markings in the diagrams.

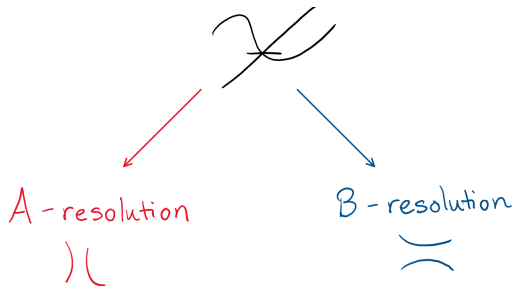


Figure: Resolving the markings of a ch-diagram

Converting $10_1^{0,1}$

The first step in converting the diagram is to isotope it into bridge position as shown here.

$10_1^{0,1}$

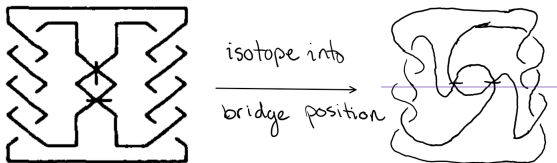


Figure: Putting the diagram into bridge position

The portion of the diagram above the bridge will give two of the diagrams when the crossings are resolved, and the portion below the bridge will give the third.

Converting $10_1^{0,1}$

Our potential triplane diagram looks like this (after a suitable planar isotopy).



Figure: Our potential tri-plane diagram

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Finally, we verify that the mirrored unions are diagrams for unlinks.

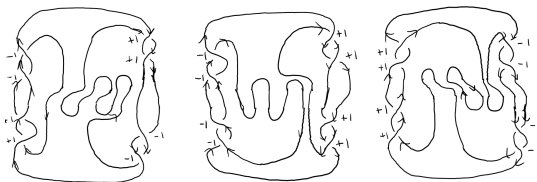


Figure: Verifying unlinks

Converting $10_1^{0,1}$

In our case, we were fortunate enough to end up with a valid tri-plane diagram. At times different bridge positions would need to be tried to develop valid diagrams.

Conclusion

Acknowledgements

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Thank you!

Questions?

Email: imartinez00@mail.fresnostate.edu